

Problem

Given that $\mathcal{F}[f(t)] = (j\omega)(e^{-j\omega} - 1)$, find the Fourier transforms of:

(a) $x(t) = f(t) + 3$

(b) $y(t) = f(t - 2)$

(c) $h(t) = f'(t)$

(d) $g(t) = 4f\left(\frac{2}{3}t\right) + 10f\left(\frac{5}{3}t\right)$

Step-by-step solution

Step 1 of 4

$$\begin{aligned} \text{(A)} \quad X(\omega) &= F(\omega) + F[3] \\ &= 6\pi\delta(\omega) + \frac{j}{\omega}(e^{-j\omega} - 1) \end{aligned}$$

Step 2 of 4

$$\begin{aligned} \text{(B)} \quad f(t) &= f(t - 2) \\ Y(\omega) &= e^{-j\omega 2} F(\omega) = \frac{je^{-2j\omega}}{\omega}(e^{-j\omega} - 1) \end{aligned}$$

Step 3 of 4

$$\begin{aligned} \text{(C)} \quad \text{if } h(t) &= f'(t) \\ H(\omega) &= j\omega F(\omega) = j\omega \frac{j}{\omega}(e^{-j\omega} - 1) \\ &= 1 - e^{-j\omega} \end{aligned}$$

Step 4 of 4

$$\begin{aligned} \text{(D)} \quad g(t) &= 4f\left(\frac{2}{3}t\right) + 10f\left(\frac{5}{3}t\right) \\ G(\omega) &= 4 \times \frac{3}{2} F\left(\frac{3}{2}\omega\right) + 10 \times \frac{3}{5} F\left(\frac{3}{5}\omega\right) \\ &= 6 \cdot \frac{j}{\frac{3}{2}\omega} \left(e^{-\frac{j3\omega}{2}} - 1 \right) + \frac{6j}{\frac{3}{5}\omega} \left(e^{-\frac{j3\omega}{5}} - 1 \right) \\ &= \frac{4j}{\omega} \left(e^{-\frac{j3\omega}{2}} - 1 \right) + \frac{j10}{\omega} \left(e^{-\frac{j3\omega}{5}} - 1 \right) \end{aligned}$$